

TC ID	Title	Module	Precondition	Steps	Expected Result	Priority	Tested By	Status
NAVIGATION, APP BOOTSTRAP, AND HOME								
TC-HOME-001	Launch app with existing PIN	Navigation, App Bootstrap, and Home	PIN already configured	Open app	PIN screen appears before Home and user cannot access Home without valid PIN	High	tester_198	Pass
TC-HOME-002	Launch app without PIN	Navigation, App Bootstrap, and Home	No PIN configured	Open app	App routes to PIN creation/setup flow before regular usage	High	tester_198	Pass
TC-HOME-003	Home screen class list loads correctly	Navigation, App Bootstrap, and Home	At least 2 classes exist	Open Home	All classes load with class name, roll prefix, and student count without reload jitter	Medium	tester_196	Pass
TC-HOME-004	Tap class without active session	Navigation, App Bootstrap, and Home	Selected class has no unfinished session	Tap class card	User navigates to Session Start screen	High	tester_198	Pass
TC-HOME-005	Tap class with active session	Navigation, App Bootstrap, and Home	Selected class has unfinished session in Room	Tap class card	Proper Session Recovery screen opens for that class only	High	tester_198	Pass
TC-HOME-006	Search/filter classes on Home	Navigation, App Bootstrap, and Home	Multiple classes exist	Search by partial class name	Matching classes filter correctly without breaking class tap navigation	Medium	tester_198	Pass
TC-HOME-007	Open Attendance folder from hamburger menu	Navigation, App Bootstrap, and Home	Attendance folder exists or app can create it	Open hamburger menu, tap Attendance folder	File manager opens exact Attendance folder instead of drifting to QuickMark root	Low	tester_198	Pass
TC-HOME-008	Open Export folder from hamburger menu	Navigation, App Bootstrap, and Home	Export folder exists or app can create it	Open hamburger menu, tap Export folder	File manager opens exact Export folder instead of drifting to QuickMark root	Low	tester_198	Pass
TC-HOME-009	Open How to Use page from hamburger menu	Navigation, App Bootstrap, and Home	App on Home	Open menu, tap How to Use	Left-side entrance animation appears and page opens without Home reload	Low	tester_196	Pass
TC-HOME-010	Open About page from hamburger menu	Navigation, App Bootstrap, and Home	App on Home	Open menu, tap About QuickMark	Left-side entrance animation appears and page opens without Home reload	Low	tester_198	Pass
AUTHENTICATION AND SECURITY								
TC-AUTH-001	Create new PIN successfully	Authentication and Security	Fresh app state without PIN	Enter valid PIN and confirm	PIN is stored and next app launch requires it	High	tester_198	Pass
TC-AUTH-002	Reject incomplete PIN on login	Authentication and Security	PIN configured	Enter fewer than required digits and confirm	Inline red error appears and stronger error haptic triggers	High	tester_198	Pass
TC-AUTH-003	Reject incorrect PIN	Authentication and Security	PIN configured	Enter wrong PIN	Access denied, error shown, error haptic triggered	High	tester_198	Pass
TC-AUTH-004	Update PIN with same value as current PIN	Authentication and Security	PIN configured	Enter correct current PIN, then set identical new PIN	Update blocked with message that new PIN must be different	Medium	tester_198	Pass
TC-AUTH-005	PIN numpad haptics work correctly	Authentication and Security	PIN screen visible	Tap digits, backspace, and confirm	Light haptic on key presses and stronger haptic on errors	Low	tester_198	Pass
TC-AUTH-006	Delete class requires valid PIN	Authentication and Security	At least one class exists	Open Delete Class, enter wrong PIN, then correct PIN	Wrong PIN blocked; correct PIN allows delete path	High	tester_198	Pass
CLASS MANAGEMENT								
TC-CLASS-001	Add class from valid QuickMark template	Class Management	Valid .xlsx template available	Open Add Class, choose file, enter class name, create class	Class imports, students parse, internal copy created, attendance folder copy created	High	tester_196	Pass

TC-CLASS-002	Reject class import with zero students	Class Management	Workbook has no valid student rows	Validate file in Add Class	Validation fails with meaningful error and Create Class remains blocked	High	tester_198	Pass
TC-CLASS-003	Reject duplicate class name	Class Management	Class already exists with same name	Try creating another class with identical class name	Create blocked with duplicate class name error	High	tester_196	Pass
TC-CLASS-004	Reject invalid roll format ending with fewer than 3 digits	Class Management	Workbook contains roll like BT24CSE01	Validate file	Validation fails with exact 3-digit roll requirement message	High	tester_196	Pass
TC-CLASS-005	Reject roll number outside 001 to 999	Class Management	Workbook contains roll ending with 000 or 1000	Validate file	Validation fails with 001 to 999 message	Medium	tester_198	Pass
TC-CLASS-006	Reject mixed roll prefixes inside same workbook	Class Management	Workbook contains at least one row with different roll prefix	Validate file	Validation fails with row-level mismatch message	Medium	tester_198	Pass
TC-CLASS-007	Reject corrupted or non-xlsx workbook	Class Management	Corrupt workbook or wrong file format	Validate file in Add Class	Validation fails with readable workbook error	High	tester_196	Pass
TC-CLASS-008	Show large-file warning for valid but heavy workbook	Class Management	Workbook exceeds large-file threshold	Validate file in Add Class	Validation succeeds, warning popup says performance may degrade	Medium	tester_196	Pass
TC-CLASS-009	Rename class successfully	Class Management	Class exists	Open Rename Class, change name, save	Class metadata updates and public workbook naming stays consistent	Medium	tester_196	Pass
TC-CLASS-010	Delete class removes metadata and files safely	Class Management	Class exists	Open Delete Class, confirm with valid PIN	Class metadata, cached students, and internal workbook are removed; export backup preserved	High	tester_198	Pass
EDIT STUDENTS								
TC-EDIT-001	Edit student name successfully	Edit Students	Class exists with students	Open Edit Students, modify name, save	Excel workbook and Room student cache update, list refreshes immediately	Medium	tester_198	Pass
TC-EDIT-002	Edit threshold with valid decimal value	Edit Students	Edit Student dialog open	Enter threshold such as 82.50 and save	Save succeeds and value persists	Medium	tester_196	Pass
TC-EDIT-003	Reject non-numeric threshold	Edit Students	Edit Student dialog open	Enter invalid threshold text	Popup error shown; save blocked	High	tester_198	Pass
TC-EDIT-004	Reject threshold outside 0.00 to 100.00	Edit Students	Edit Student dialog open	Enter threshold less than 0 or greater than 100	Popup error shown; save blocked	High	tester_198	Pass
TC-EDIT-005	Toggle inactive flag successfully	Edit Students	Edit Student dialog open	Toggle inactive flag and save	Workbook and Room update correctly	Medium	tester_196	Pass
TC-EDIT-006	Block duplicate student name in same class	Edit Students	Another student in same class already has target name	Change selected student name to duplicate and save	Save blocked with popup error	High	tester_198	Pass
TC-EDIT-007	Save with no changes	Edit Students	Edit Student dialog open	Press Save without changing name, inactive flag, or threshold	No Excel write, no Room write, inline red warning says Nothing changed	Low	tester_196	Pass
SESSION START AND SESSION RECOVERY								
TC-REC-001	Start new roll call session from Session Start	Session Start and Session Recovery	Class exists and no active session for that class	Open Session Start, choose roll-call path, start session	Room session buffer created and Roll Call opens	High	tester_196	Pass
TC-REC-002	Start MPS flow from Session Start	Session Start and Session Recovery	Class exists and no active session for that class	Open Session Start, choose Mark Present / Absent Students	MPS screen opens with correct class context	High	tester_196	Pass

TC-REC-003	Set continuous session count from dropdown	Session Start and Session Recovery	Session Start open	Select count from 1 to 10 and start session	Selected count persists in active session metadata	Medium	tester_198	Pass
TC-REC-004	Recovery screen shows correct class-specific unfinished session	Session Start and Session Recovery	Multiple classes have unfinished sessions	Tap class A, then tap class B from Home	Recovery screen shows matching class, last marked number, and stage for each tapped class only	High	tester_196	Pass
TC-REC-005	Recovery discard requires confirmation once	Session Start and Session Recovery	Session Recovery screen open	Tap Discard Session and confirm	Single confirmation popup shown; session clears only after confirmation	Medium	tester_198	Pass
TC-REC-006	Recovery continue to roll call	Session Start and Session Recovery	Session stage is attendance	Tap Continue Attendance	Roll Call opens and back from resumed flow returns to Home, not stale recovery stack	High	tester_196	Pass
TC-REC-007	Recovery continue to MPS	Session Start and Session Recovery	Session stage is mark present / absent	Tap Continue via MPS	MPS opens with persisted state and correct last-tapped context	High	tester_198	Pass
TC-REC-008	Recovery continue to manual attendance	Session Start and Session Recovery	Session stage is manual attendance	Tap Continue via Manual Attendance	Manual Attendance opens and recovery loop remains stable across repeated passes	High	tester_196	Pass
TC-REC-009	Recovery continue to summary	Session Start and Session Recovery	Session stage is summary	Tap Continue To Summary	Summary opens directly with summary-stage recovery behavior	High	tester_196	Pass
ROLL CALL AND TTS								
TC-RC-001	First roll blocks buttons until spoken when TTS enabled	Roll Call and TTS	Roll Call opens with TTS on	Enter Roll Call and observe Present/Absent buttons	Buttons remain blocked until first roll speech actually starts	High	tester_198	Pass
TC-RC-002	No TTS mode skips speech wait	Roll Call and TTS	TTS mode set to No TTS	Start roll call	No speech; buttons available without waiting for utterance start	High	tester_198	Pass
TC-RC-003	Turning TTS on from Roll Call while currently off	Roll Call and TTS	Roll Call open with No TTS selected	Change dropdown from No TTS to a real speed option	Selected voice invokes current roll speech and buttons stay blocked until speech starts	Medium	tester_198	Pass
TC-RC-004	Mark present and absent responses advance correctly	Roll Call and TTS	Roll Call active	Tap Present on one roll and Absent on next roll	Session attendance updates and next valid roll displays	High	tester_198	Pass
TC-RC-005	Last marked number tracks lecturer response, not last spoken roll	Roll Call and TTS	Roll Call active	Respond to a roll, exit, reopen recovery	Recovery shows last responded roll number only	High	tester_196	Pass
TC-RC-006	Re-edit previous roll after reaching final student	Roll Call and TTS	Roll Call reaches end of range	Navigate back to prior roll and mark again	Buttons remain usable for attendance correction	Medium	tester_198	Pass
TC-RC-007	Exit popup on main roll-call flow with zero marked students	Roll Call and TTS	Fresh roll-call session, no student marked yet	Tap back	Exit popup hides Save and shows only valid options	Medium	tester_198	Pass
TC-RC-008	Exit popup on resumed roll-call flow	Roll Call and TTS	Roll Call opened via recovery	Tap back and choose Discard	Confirmation rewrite appears for recovery-continued discard path	Medium	tester_198	Pass
TC-RC-009	TTS prewarm on Session Start and Recovery continue	Roll Call and TTS	TTS enabled	Start session from Session Start, then continue later from Recovery	Roll Call engine is warmed before screen use; first speech delay reduced compared with cold start	Low	tester_196	Pass
MPS (MARK PRESENT SCREEN)								
TC-MPS-001	MPS opens with correct title and state behavior	MPS (Mark Present Screen)	Session Start or Recovery opens MPS	Open MPS	Screen shows MPS layout and marks/restores selected students correctly	High	tester_196	Pass
TC-MPS-002	Mark student present from MPS	MPS (Mark Present Screen)	MPS open with selectable student grid	Tap a student chip	Student marked present and state persists in session buffer	High	tester_196	Pass

TC-MPS-003	Change MPS-selected student status	MPS (Mark Present Screen)	MPS open with one previously marked student	Tap a student chip	Student state changes correctly and persists	Medium	tester_196	Pass
TC-MPS-004	Resume MPS from recovery with persisted state	MPS (Mark Present Screen)	MPS session partially completed and exited	Reopen class from Home and continue via MPS	Previous marks and last tapped location restore correctly	High	tester_198	Pass
TC-MPS-005	Exit MPS to Home using Save or Discard	MPS (Mark Present Screen)	MPS active	Tap back and choose Save or Discard	User goes to Home, not Session Start	Medium	tester_196	Pass
MANUAL ATTENDANCE								
TC-MAN-001	Manual Attendance shows only remaining unmarked roll numbers	Manual Attendance	Some students already marked in session	Open Manual Attendance	Only remaining unmarked roll numbers appear in increasing order	High	tester_196	Pass
TC-MAN-002	Mark individual student present or absent	Manual Attendance	Manual Attendance open with remaining rolls shown	Tap P for one student and A for another	Marks persist immediately to session storage	High	tester_198	Pass
TC-MAN-003	Bulk mark remaining students present	Manual Attendance	Manual Attendance open with remaining unmarked students	Tap Mark All Present	All remaining students marked present	Medium	tester_196	Pass
TC-MAN-004	Bulk mark remaining students absent	Manual Attendance	Manual Attendance open with remaining unmarked students	Tap Mark All Absent	All remaining students marked absent	Medium	tester_198	Pass
TC-MAN-005	Prevent final save while any remaining students are unmarked	Manual Attendance	At least one student still unmarked	Tap Save Attendance	Save blocked and user gets mark-all-students-first message	High	tester_198	Pass
TC-MAN-006	Manual save from exit popup when zero manual marks exist	Manual Attendance	Enter Manual Attendance and mark nothing	Tap back, choose Save	User goes Home; tapping same class later opens proper full recovery screen	Medium	tester_198	Pass
TC-MAN-007	Manual save from exit popup after manual marks exist	Manual Attendance	Mark at least one manual attendance value	Tap back, choose Save	Session remains on manual-attendance recovery branch with correct last marked number	Medium	tester_198	Pass
TC-MAN-008	Manual recovery loop remains stable on repeated passes	Manual Attendance	Unfinished session exists	Recovery -> Manual -> Save/Recovery -> repeat several times	First recovery always opens proper Session Recovery screen, not Home accidentally	High	tester_196	Pass
SUMMARY AND RESULT								
TC-SUM-001	Summary shows present and absent sections with 3-digit roll chips	Summary and Result	Session has marked students	Open Summary	Present and absent sections show correct students with 3-digit roll chip format	High	tester_196	Pass
TC-SUM-002	Update attendance writes final session to Excel	Summary and Result	Summary ready with final data	Tap final update button	Excel write completes, temporary session state clears, Result screen opens	High	tester_196	Pass
TC-SUM-003	Summary back double-press behavior	Summary and Result	Summary open	Press back once, then press back again within allowed interval	First back warns; second back exits to Home while preserving summary-stage recovery behavior	Medium	tester_198	Pass
TC-SUM-004	Result screen back navigation	Summary and Result	Result screen open	Use system back	User goes to Home, not back into Summary	Medium	tester_196	Pass
EXCEL VALIDATION AND WORKBOOK SAFETY								
TC-XLS-001	Validate exact QuickMark headers	Excel Validation and Workbook Safety	Workbook header modified incorrectly	Import in Add Class	Validation fails with exact cell mismatch message	High	tester_198	Pass
TC-XLS-002	Validate Attendance (%) formula for active rows	Excel Validation and Workbook Safety	Active student row formula altered	Import workbook	Validation fails with Attendance (%) formula mismatch	High	tester_198	Pass

TC-XLS-003	Validate threshold conditional formatting rule	Excel Validation and Workbook Safety	Column E warning conditional rule missing or wrong	Import workbook	Validation fails with threshold conditional formatting error	High	tester_198	Pass
TC-XLS-004	Validate inactive conditional formatting rules	Excel Validation and Workbook Safety	Inactive styles changed in workbook	Import workbook	Validation fails for inactive flag/name/attendance formatting mismatch	High	tester_196	Pass
TC-XLS-005	Validate inactive attendance display as N/A	Excel Validation and Workbook Safety	Inactive row displays anything other than N/A in Attendance (%)	Import workbook	Validation fails with inactive attendance display error	High	tester_196	Pass
TC-XLS-006	Validate threshold range in workbook rows	Excel Validation and Workbook Safety	Row threshold outside 0 to 100	Import workbook	Validation fails with row-specific threshold range error	High	tester_196	Pass
EXCEL WRITE, SESSION PERSISTENCE, AND RECOVERY RESILIENCE								
TC-WRITE-001	Final write preserves workbook format	Excel Write, Session Persistence, and Recovery Resilience	Valid class workbook and completed summary	Finalize attendance	New session column, styles, formulas, and formatting remain intact	High	tester_198	Pass
TC-WRITE-002	Continuous session write stores exact chosen session span	Excel Write, Session Persistence, and Recovery Resilience	Session started with continuous count greater than 1	Finalize attendance	Present values write exact selected count; absent writes 0; header carries cN suffix	High	tester_198	Pass
TC-WRITE-003	Session data survives app switch during attendance	Excel Write, Session Persistence, and Recovery Resilience	Attendance session active	Switch apps and return	Session remains usable or recoverable without data loss	High	tester_196	Pass
TC-WRITE-004	Session data survives app kill mid-session	Excel Write, Session Persistence, and Recovery Resilience	Attendance session active	Kill app and reopen	Tapped class shows recoverable unfinished session path	High	tester_198	Pass
TC-WRITE-005	Multiple unfinished sessions across classes remain isolated	Excel Write, Session Persistence, and Recovery Resilience	Two or more classes have unfinished sessions	Start and leave unfinished sessions in multiple classes	Each class recovers its own active session without collision	High	tester_196	Pass
EXPORT AND PDF REPORTING								
TC-EXP-001	Export Excel workbook to export folder	Export and PDF Reporting	Class exists	Open Export screen and choose Excel export	Workbook copy saved in QuickMark Exports folder	Medium	tester_196	Pass
TC-EXP-002	Export PDF roster successfully	Export and PDF Reporting	Class exists	Open Export screen and choose PDF export	PDF generated in QuickMark Exports folder	High	tester_196	Pass
TC-EXP-003	Block navigation during export	Export and PDF Reporting	Export in progress	Start export and immediately try to go back	Navigation remains blocked until export completes	Medium	tester_198	Pass
TC-EXP-004	PDF shows attendance percentage values	Export and PDF Reporting	Workbook contains attendance percentage values or derivable session values	Generate PDF	Attendance (%) column is visible, not blank	High	tester_196	Pass
TC-EXP-005	PDF threshold coloring	Export and PDF Reporting	Students above and below threshold exist	Generate PDF roster	Attendance percentage above threshold is green; below threshold uses warning color	Medium	tester_198	Pass
TC-EXP-006	PDF multi-session date labels on same day	Export and PDF Reporting	Same-day multiple session columns exist	Generate PDF	Date headers preserve suffixes such as 18 and 18(1)	Medium	tester_198	Pass
SETTINGS, TTS PREFERENCES, AND SUPPORT PAGES								
TC-SET-001	Change TTS speed in Settings	Settings, TTS Preferences, and Support Pages	Settings open	Select a new TTS speed	Preference persists and Roll Call uses updated speed	Medium	tester_196	Pass
TC-SET-002	Select No TTS in Settings	Settings, TTS Preferences, and Support Pages	Settings open	Select No TTS	Voice test disabled and Roll Call skips speech waiting behavior	High	tester_196	Pass

TC-SET-003	Re-enable TTS after No TTS	Settings, TTS Preferences, and Support Pages	TTS currently disabled	Select valid TTS option in Settings	Voice option and roll-call speech return correctly	Medium	tester_196	Pass
TC-SET-004	Open How to Use, About, and Excel Colors support pages	Settings, TTS Preferences, and Support Pages	Home menu available	Open each support page	Each screen opens with correct animation and back behavior without reloading Home	Low	tester_196	Pass
FOLDER, FILE, AND APP-STORAGE BEHAVIOR								
TC-FILE-001	Attendance and Export folders recreated safely if missing	Folder, File, and App-Storage Behavior	Public folder deleted externally	Trigger action that uses folder	Folder recreated or write path handled safely	High	tester_198	Pass
TC-FILE-002	Clear app storage behavior awareness	Folder, File, and App-Storage Behavior	App storage cleared manually	Reopen app	App behaves like fresh install; Room, prefs, and internal files are gone; public folders may remain	Medium	tester_198	Pass
TC-FILE-003	Internal class copy exists immediately after Add Class	Folder, File, and App-Storage Behavior	Valid class imported	Add class and inspect attendance flow readiness before first session	Internal workbook and attendance-folder copy already exist before any session write	High	tester_196	Pass